VO ROBOTICS · DOCK PROJECTS

Dock Projects

Two beginner-friendly projects on our robot charging dock. You need basic Python for Project 1 and an interest in electronics for Project 2. You will learn the rest as you go. **You work from home on your own laptop. No hardware is needed.**

What is the dock?

Our robots drive onto a **charging dock** to charge their batteries. The dock is a small box with:

- **Two metal pads** on the floor. The robot's charging contacts touch them.
- **A charger** (29.4 V) that charges the robot's battery.
- **A relay**, an electric switch that connects the charger to the pads only when a robot is there.
- **A voltage sensor** that measures the voltage on the pads.
- **A small computer (ESP32-S3)** with Wi-Fi that controls everything and shares its status as data on the network.

Words you will meet

Word	Simple meaning
Voltage (V)	The electrical "pressure". A full robot battery is about 29 V.
Current (A)	How much electricity is flowing, measured in amps.
Relay	A switch that a small signal can turn on and off.
API	A web address that returns data for programs instead of a web page for people.
JSON	A simple text format for data, like `{"state": "CHARGING", "pad_v": 28.9}`.
CSV	A table saved as text, one row per line. Opens in Excel.
Charge session	One visit of a robot: from arriving on the dock until it leaves.
Simulator	A program that pretends to be the real dock, so you can test without hardware.
Datasheet	The official PDF from a manufacturer with a part's limits and details.

Safety rules (read first)

- Your programs may only **read** data from the dock. They never send commands or change settings.
- Work only with the files we send you. You do not need a real dock or any parts, and you do not need to buy anything.
- When you write a number in a report, say where it came from: a datasheet page, a recording or a simulation.
- Save your work in Git every day. Ask your supervisor when you are stuck.

PROJECT 1 · PYTHON

Dock Recorder

Goal: write a Python program that records what the dock is doing every second, and then makes a short report of each charge session.

Why it helps us: when a dock has a problem, we need to see exactly what happened and when. By hand this takes hours; your program will do it in minutes.

You will get: a zip file with two real recordings from our dock and a **dock simulator**. The simulator runs on your laptop and behaves like the real dock: same web addresses, same data. Start it with:
`python dock_simulator.py data/recording_B_evening.csv`

Steps

Step 1 Open `http://localhost:8080/api/status` in your browser. You will see JSON. Find these fields:

Field	Meaning
`state`	What the dock is doing: IDLE, APPROACH, STABILIZING, CHARGING, CHARGED, UNDOCKING, BAD_CONTACT or FAULT
`pad_v`	Voltage on the pads, in volts
`relay_closed`	`true` when the charger is connected to the pads
`robot_present`	`true` when a robot battery is seen on the pads
`fault`	Error text, empty when everything is fine

Step 2 Write a Python script that reads `/api/status` every 1 second and adds one row to a CSV file: time, state, pad_v, relay_closed, robot_present.

Step 3 Make it robust: if Wi-Fi drops or the dock restarts, your program must not crash. It should wait, try again, and note the gap in the file. Test this with the simulator's problem mode: `--drops 0.05 --restart-every 5`. Tip: when `uptime_ms` goes down, the dock has restarted.

Step 4 Find the charge sessions in your CSV. A session starts when `state` becomes CHARGING and ends when it is no longer CHARGING. For each session, write: start time, end time, duration in minutes, start voltage, end voltage.

Step 5 Draw one chart per session (voltage over time) and write a short report with a table of all sessions.

Bonus: warn when a charge lasts less than 1 minute; also read `/api/events` (recent dock messages, such as `"Relay CLOSED"`); record several docks at once (run two simulators with `--port 8080` and `--port 8081`).

Helpful Python libraries: `requests` (read the web address), `csv`, `time`, `datetime`, `matplotlib` (charts), `pandas` (optional, for tables).

What to hand in

- Your code in a Git repository, with a README that explains how to run it.
- The CSV files and charts from both recordings, and from one 2-hour run on the simulator with problem mode on.
- One page in your own words: what did you see in the data?

How we will mark it

What we check	Marks
It works: the sessions and times match what really happened	**40**
It does not crash in problem mode; it runs for 2 hours without stopping	**20**
Your one-page report explains the data clearly	**20**
Clean code and a README that a classmate can follow	**20**

Pass mark: 60 out of 100.

PROJECT 2 · ELECTRONICS RESEARCH

Dock Circuit Research

Goal: the dock works, but some of its circuits can be improved. For five parts of the dock, find two or three possible circuits, compare them using datasheets, and recommend the best one.

Why it helps us: your final table will help the team choose the circuits for the next version of the dock. Nothing changes on the real dock until the team decides.

This is desk research: everything is done online with datasheets and shop pages. You do not buy, build or measure anything.

How to research one part

1. **Understand the problem.** Read what we need (below) and ask questions.
2. **Find options.** Search for 2–3 different circuits or ready-made modules. Use the search words given.
3. **Read the datasheets.** Note the maximum voltage, maximum current, price and where to buy.
4. **Compare.** Fill in a table like the one below.
5. **Choose.** Pick the best option and explain why in 2–3 sentences.

The five parts

Part	What we need
A. Measuring current	Today the dock only measures voltage. We want it to also measure how much current flows into the robot. Search: "INA228 shunt resistor", "ACS712 current sensor", "current sense amplifier"
B. Charger switch	A safe way to connect and disconnect the charger. It must switch **off** if the controller restarts, and it must work from a 3.3 V signal. Search: "relay driver transistor circuit", "MOSFET switch 30 V", "3.3 V relay module optocoupler"
C. Protecting the pads	Protection against short circuits, wrong polarity and voltage spikes. The pads are outdoors and metal. Search: "fuse for 30 V DC", "reverse polarity protection MOSFET", "TVS diode 33 V"
D. Power for the controller	A steady 5.1–5.2 V supply, so that the controller starts correctly every time and its Wi-Fi does not drop. Search: "5 V DC-DC buck module", "voltage supervisor reset IC", "ESP32 boot pin pull-up"
E. Detecting the robot	A second way to know the robot is on the dock, without using its battery voltage. It must work in sand, dust and sun. Search: "hall sensor magnet detection", "IR proximity sensor module", "limit switch waterproof"

Comparison table (one per part)

Option	Part name	Max voltage	Max current	Price	Where to buy	Good points	Bad points
1							
2							
3							

Add the link to each datasheet. If you cannot find a value, write "not found" rather than guessing.

Final table: your recommendations

Part	My choice	Backup choice	Why	Price
A. Measuring current				
B. Charger switch				
C. Protecting the pads				
D. Power for the controller				
E. Detecting the robot				

You will present this table to the team. The team then marks each part "agreed", "change" or "research more".

What to hand in

- A short report or slides with your comparison tables and datasheet links.
- Your final recommendations table.
- Optional: a simple circuit drawing or a simulation in a free online tool (for example Falstad or Tinkercad Circuits).

How we will mark it

What we check	Marks
Every number comes from a datasheet, with a link	**30**
Your choices meet the needs (voltage, current, safety)	**25**
Tables are clear and easy to compare	**25**
Presentation is clear	**20**

Pass mark: 60 out of 100, with parts A and B complete.

Questions? Ask your supervisor.